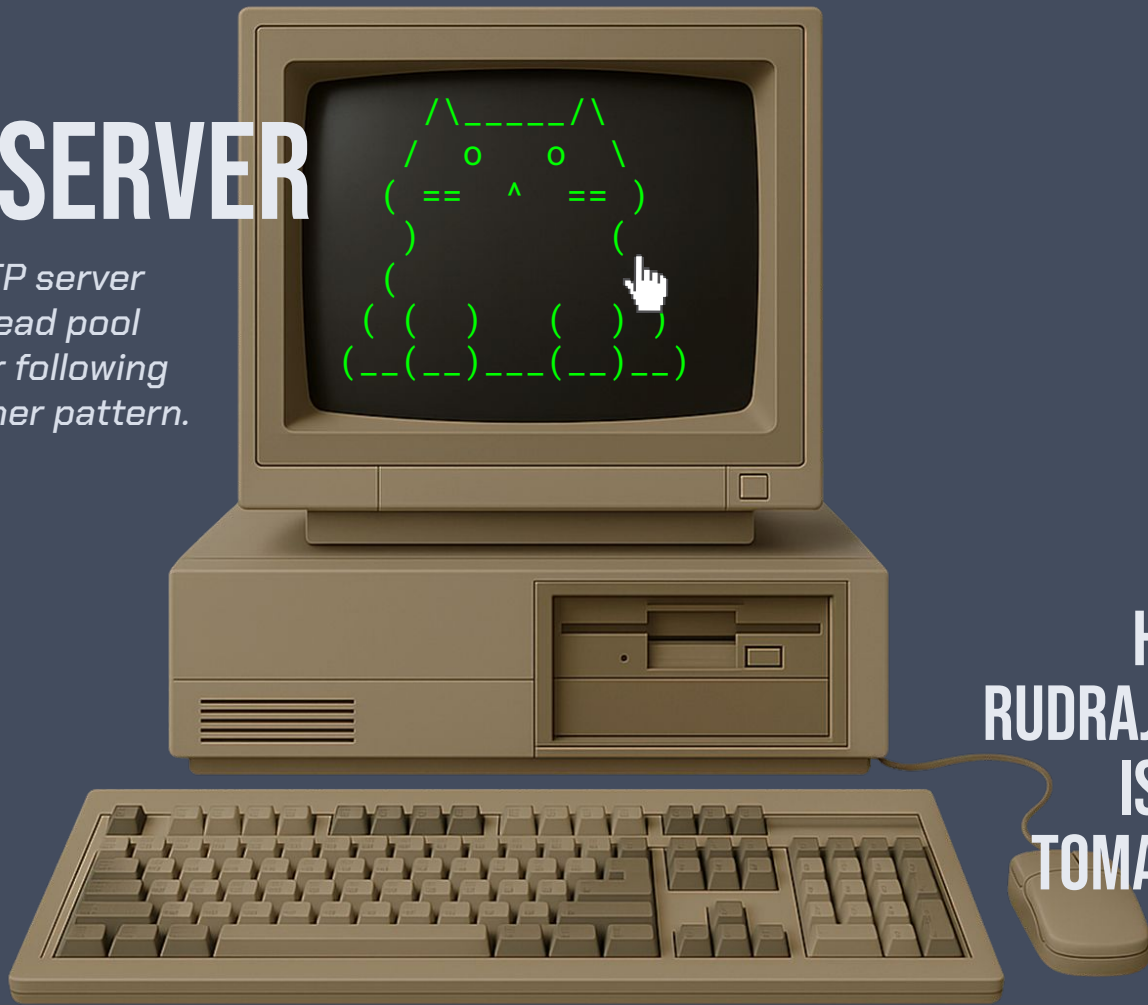


TEAM 0101

G.5 WEB SERVER

*A multi-threaded HTTP server
using a fixed-size thread pool
and a bounded buffer following
the producer-consumer pattern.*



HELENA ZHUO
RUDRAJIT BANERJEE
ISHIT ARHATIA
TOMAS COLORADO

INTRODUCTION:

concurrent web servers

```

{ "request": "Churu" }
  |
  |
  |
{ "request": "pets" }
  \   |   /
   / \   \
  /   o   o \
 /== ^ ==\
(   )   (   )
(   )   (   )
( ( )   ( ( ) )
(---)---(---)---
  
```

printf("mrow!!")

```

{ "request": "kibble" }
  |
  |
  |
{ "request": "chompArm" }
  
```

INTRODUCTION:

architecture & theory

```
{ buffer_put(bounded_buffer_t *buf,  
            request_t req) } !!!
```

mrow i'm starving! :(



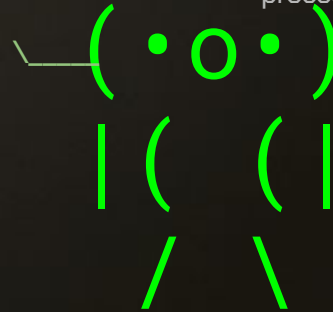
bounded buffer

threads..

threads...

work.. work.. process..

process..



INTRODUCTION:

contemporary systems

NGINX

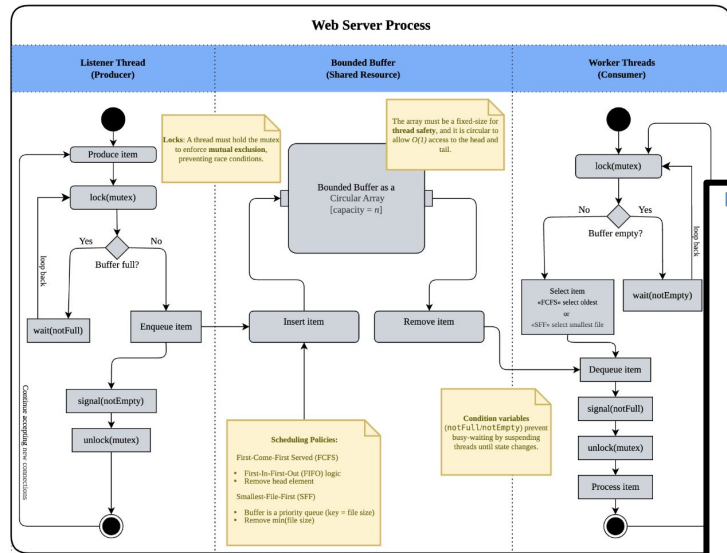
Apache HTTP

(o_o) (•_—)
\\ ((> <)) J /\\)
(/) ()
\\(--)| / \\ | (--) /

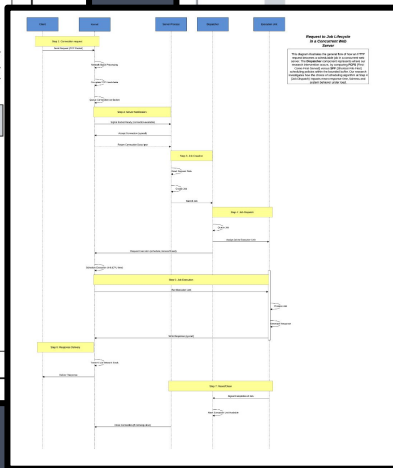
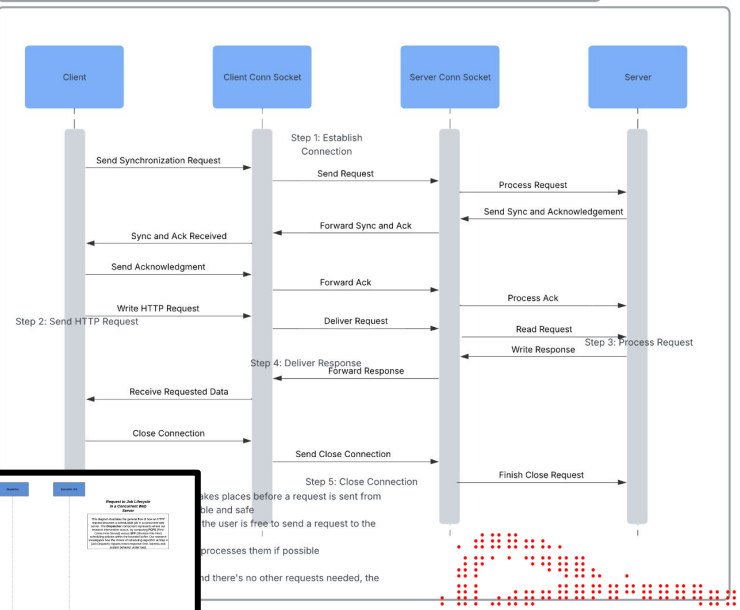
IMPLEMENTATION DESIGN: UML DIAGRAMS

Bounded Buffer Activity Diagram

This UML diagram demonstrates the **critical path** of a request in a multi-threaded web server, as every request passes through the **bounded buffer**, interacting with listener and worker threads. This critical path is often a bottleneck that determines server performance and efficiency, and is significantly affected by the scheduling policies that our research extension into First-Come-First-Serve and Shortest-File-First explores.



HTTP Request-Response Sequence Diagram



PROBLEM:

workloads of varying sizes



PROBLEM:

key performance indicators

throughput

ooooooooo ✓ 32,540 requests handled!



0.004 nanosec

transfer rate 84156.3
time per request 0.127

mean response time

Running 30s test @ http://localhost:8080

2 threads and 25 connections

Thread Stats	Avg	Stdev	Max	+/-	Stdev
Latency	2.33ms	1.57ms	33.03ms	81.21%	
Req/Sec	5.00k	780.15	6.39k	76.33%	

Latency Distribution

50%	2.06ms
75%	2.91ms
90%	3.94ms
99%	8.08ms

298698 requests in 30.03s, 30.31GB read

Requests/sec: 9945.82

Transfer/sec: 1.01GB

```
licensed to The Apache Software Foundation, http://www.apache.org/

enchmarking localhost (be patient)
ompleted 10000 requests
ompleted 20000 requests
ompleted 30000 requests
ompleted 40000 requests
ompleted 50000 requests
ompleted 60000 requests
ompleted 70000 requests
ompleted 80000 requests
ompleted 90000 requests
ompleted 100000 requests
inished 100000 requests

erver Software: Apache/2.4.52
erver Hostname: localhost
erver Port: 80

ocument Path: /
ocument Length: 10671 bytes

oncurrency Level: 500
ime taken for tests: 12.699 seconds
omplete requests: 100000
ailed requests: 0
otal transferred: 1094500000 bytes
ML transferred: 106710000 bytes
equests per second: 7874.40 [#/sec] (mean)
ime per request: 63.497 [ms] (mean)
ime per request: 0.127 [ms] (mean, across all concurrent requests)
ransfer rate: 84165.38 [Kbytes/sec] received

onnection Times (ms)
min mean +/- sd median max
```


SOLUTION:

SIZE-BASED SCHEDULING

- 1  = [. . .] = a small mrow 4kb
- 2  = [. . .] = a big mrow 1MB!! -- get in line, felines.
- 3  = [. . .] = a small mrow 4kb
- 4  = [. . .] = a medium mrow.. between 4kb and 1MB

FCFS

First-Come-First-Served

smallest mrows first! --

- 1  = [. . .] = a small mrow 4kb
- 2  = [. . .] = a small mrow 4kb
- 3  = [. . .] = a small mrow 4kb
- 4  = [. . .] = a big mrow 1MB!!

SFF

Shortest-File-First

VS

CONCURRENT WEB SERVER:

implementation - core

```
src/
├── main.c           # Entry point
├── bounded_buffer.c # mutex, condvars,
                    # FCFS/SFF dequeue
├── thread_pool.c    # init, worker creation
└── http.c           # parsing, status
                    # codes, file send
```

Team 0101's Concurrent Web Server

By Helena Zhuo, Rudrajit Banerjee, Ishit Arhatia, Tomas Colorado

A multi-threaded HTTP server using a fixed-size thread pool and a bounded buffer, following the producer-consumer pattern.

See our Github for more information: [README.md](#) [Research Extension](#) [Glossary.md](#) [Benchmarking Rig](#) [Bibliography](#)

Workload Simulator

Simulate a Workload

```
./server
```

```
===== CONFIGURATION =====
```

```
Port: 8080
```

```
Threads: 4
```

```
Buffer Capacity: 16
```

```
Scheduling Policy: FCFS
```

```
Thread Pool Initialization: 4 workers spawned
```

```
===== SERVER RUNNING =====
```

```
Ctrl+click [cmd+click on Mac] to open in your browser: http://localhost:8080
```

implementation - core [continued]

CONCURRENT WEB SERVER: IMPLEMENTATION - RESEARCH EXTENSION

*welcome to class.
this is very important.*

*^_^
(o.o)
> ^ <
/ ~ \
(_/---\)*

STAT()

USE THIS TO CHECK EACH FILE SIZE.

*ALSO, TO MAKE SURE U REALLY GET IT,
REWRITE THIS ENTIRE PROGRAM FOR HOMEWORK.
THEN 3 OTHER UNIX FUNCTIONS TOO.*

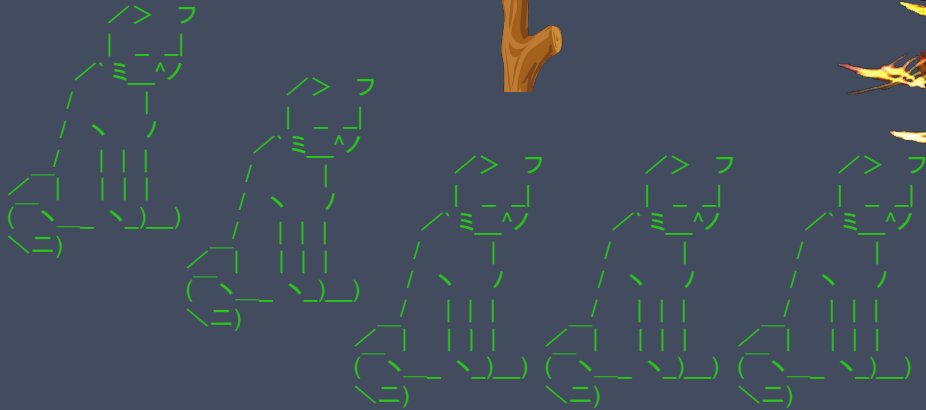
CONCURRENT WEB SERVER: IMPLEMENTATION - RESEARCH EXTENSION

this way to critical section

THE MUTEX INTO OBLIVION

*this is the critical
section*

*get your
req = buffer_get_sff(buf)
here*



IMPLEMENTATION: DESIGN CHOICES

CONFIGURABILITY

RUNTIME FLAGS



port

default: 8080



thread count

default: 4



buffer size

default: 16



scheduling policy

default: FCFS

MEASUREMENTS: METHODOLOGY & SOURCES

POLLING

Welcome to the official GNU Netcat project homepage

The GNU Netcat project



LOAD GENERATOR

NETCAT, WRK & LUA



Will Glozer
wg



wrk

Public

Modern HTTP benchmarking tool



C



40.2k



3k



MEASUREMENTS: ENVIRONMENT & DESIGN

rigor & reproducibility



VM SETUP

- ORACLE VIRTUALBOX
- UBUNTU 22.04 LTS
- 4 VCPUS, 8 GB RAM
- GCC 11.4
- WRK 4.1.03



ANALYSIS: PERFORMANCE

Workload	Concurrency	FCFS	SFF	Gain
Small (4 KB)	10	0.56	0.56	~0%
	100	3.55	4.10	-15.5%
Large (1 MB)	10	6.64	5.84	+12.0%
	100	56.34	60.66	-7.7%
Heavy-tailed	10	1.21	1.13	+6.6%
	100	8.79	10.00	-13.8%

ANALYSIS: PERFORMANCE - WHY?

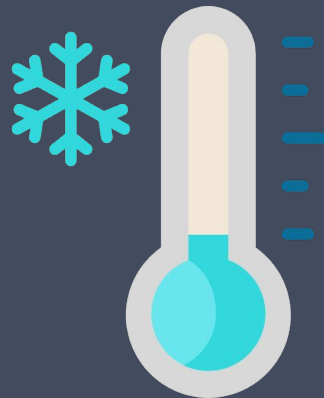
3 MAIN FACTORS!



THE SFF SCAN
(OVERHEAD)



ALWAYS BUSY WORKER
THREADS...
(THREAD SATURATION)



SERVER RESTART
VARIANCE
(COLD-START CACHE STATE)

ANALYSIS: IMPACT OF SCHEDULING POLICIES

Yeah, and only under low currency.. we don't
even want to talk about under high
contention...

SFF

\

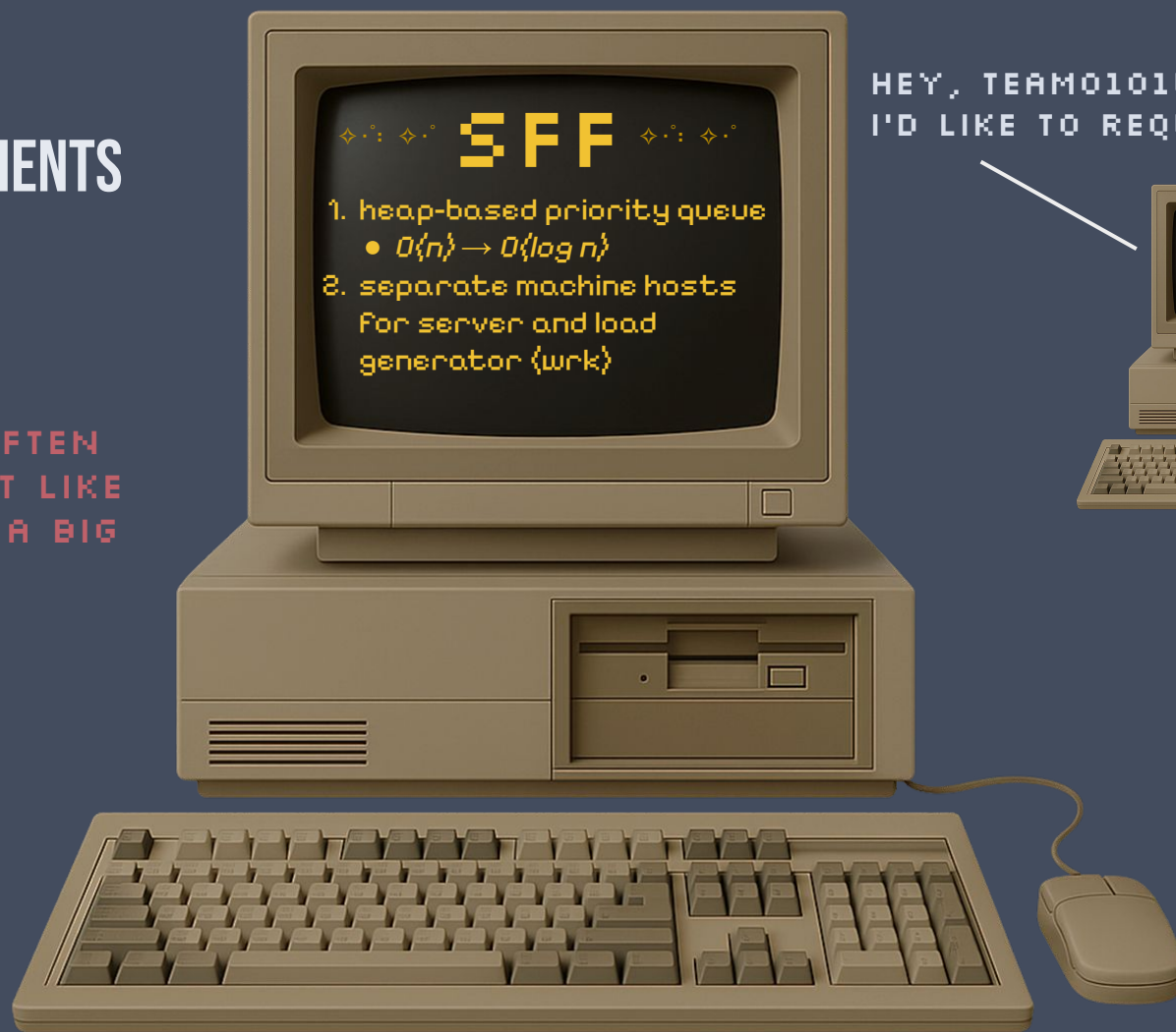
The improvements are...
MARGARINE-nally beneficial...



FOFS

CONCLUSION & FUTURE IMPROVEMENTS

IT'S NOT OFTEN
THAT A CAT LIKE
ME MAKES A BIG
REQUEST..



HEY, TEAM0101WEBSERVER.
I'D LIKE TO REQUEST(FILE).

